

Formal Analysis of Kang et al.'s Authentication Protocol using Tamarin–Prover

Publisher: IEEE

[Cite This](#)[Alsita Putri Iriana](#) ; [Septha Siswantyo](#) [All Authors](#)2
Cites in
Papers232
Full
Text Views

Abstract

Document Sections

- I. Introduction
- II. Theoretical Background
- III. Methodology
- IV. Protocol Analysis
- V. Conclusion

Authors

Figures

References

Citations

Keywords

Metrics

More Like This

Abstract:

The increasing use of IoT leads to additional IoT cloud data and also security and privacy issues. Therefore, it is necessary to have an authentication protocol on the IoT cloud. One of the authentication protocols proposed for cloud IoT is an improved version of Amin et al.'s protocol, Kang et al.'s authentication protocol. The protocol combines the pseudo-identity and actual identity of the cloud server or user with the secret number owned by control server to prevent impersonation attacks effectively. However, no formal analysis has been carried out to prove the strength of the protocol. This research proposed a formal analysis using Tamarin–Prover, which is one of the tools to model a protocol symbolically and analyze the protocol formally. The research steps were performed based on the protocol analysis methodology. Verification using Tamarin–Prover is carried out on secrecy and authentication aspects, including aliveness, weak agreements, non-injective agreements, and injective agreements. The verification results show that Kang et al.'s authentication protocol satisfies the secrecy aspect for each entity's nonce and session key, as well as the authentication aspect for the authentication process carried out by the control server to the user, the control server to the cloud server, the cloud server to the control server, and the user to the cloud server.

Published in: [2021 6th International Workshop on Big Data and Information Security \(IWBIS\)](#)

Date of Conference: 23–25 October 2021

DOI: [10.1109/IWBIS53353.2021.9631852](#)Date Added to IEEE *Xplore*: 28 December 2021

Publisher: IEEE

✓ ISBN Information:

Conference Location: Depok, Indonesia

401 Unknown Reason

Unknown Reason

This server could not verify that you are [Sign in to Continue Reading](#) ment requested. Either you supplied the wrong credentials (e.g., bad password), or your browser doesn't understand how to supply the credentials required.

Authors



Figures



References



Citations



Keywords



Metrics



Need
Full-Text
access to IEEE *Xplore*
for your organization?

CONTACT IEEE TO SUBSCRIBE >

IEEE Personal Account

CHANGE
USERNAME/PASSWORD

Purchase Details

PAYMENT OPTIONS
VIEW PURCHASED
DOCUMENTS

Profile Information

COMMUNICATIONS
PREFERENCES
PROFESSION AND
EDUCATION
TECHNICAL INTERESTS

Need Help?

US & CANADA: +1 800
678 4333
WORLDWIDE: +1 732 981
0060
CONTACT & SUPPORT

Follow



[About IEEE *Xplore*](#) | [Contact Us](#) | [Help](#) | [Accessibility](#) | [Terms of Use](#) | [Nondiscrimination Policy](#) | [IEEE Ethics Reporting](#)  | [Sitemap](#) | [IEEE Privacy Policy](#)

A public charity, IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.

© Copyright 2026 IEEE – All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies.